HANNAH FITZSIMMONS

Full Stack Software Engineer

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PROFESSIONAL EXPERIENCE

Northrop Grumman
Principal Web Software Developer *Oct 2024 – Present*

- Led full-stack development across 700+ deliverables spanning 3+ enterprise applications, including 232 stories and 283 bug resolutions over 3.5 years.
- Designed and implemented authentication/authorization architecture including identity provider integration, RBAC controls, error handling, and audit logging — with comprehensive documentation for long-term maintainability.
- Led architectural migration from client-side to server-side configuration management, implementing persistent storage and RESTful APIs for an enterprise application, improving reliability and scalability.
- Delivered 15+ production features across 3 sprints, including real-time data visualizations, connection status monitoring, and tactical reporting systems.
- Implemented reusable component libraries and enforced UI consistency and accessibility standards across multiple applications.
- Enhanced application performance through code-splitting, lazy loading, and dependency optimization.
- Mentored junior developers in front-end best practices, accessibility standards, and Agile workflows.

Northrop Grumman
Web Software Developer *Apr 2023 – Oct 2024*

- Architected authentication and session management systems, resolving critical security vulnerabilities including double-login bugs and implementing idle timeout with token expiration guards.
- Built 15+ major feature systems including initiative management (checklists, metrics, time tracking), BOM/kitting workflows, post-release order editing, and real-time data visualization dashboards.
- Refactored critical architecture to eliminate cascading failures using defensive programming patterns, custom React hooks with proper lifecycle management, and event-driven state updates.
- Resolved production-critical bugs including routing issues, application hangs, and authentication vulnerabilities through systematic debugging and root cause analysis.
- Implemented complex domain logic for calibration tracking, shelf-life management, part serialization, and conditional field behavior across interconnected systems.

Northrop Grumman
Associate Web Software Developer *Oct 2022 – Apr 2023*

- Built UI/UX features including iterative mockup design, multi-select grids, dynamic banners, tab management, and conditional form fields with stakeholder feedback integration.
- Resolved critical data conversion and display bugs including timestamp formatting, auto-refresh issues, and dynamic column visibility based on business rules.
- Contributed to extensive bug resolution efforts through automated testing, detailed reproduction steps, and console error analysis.
- Authored and refined internal documentation, including version control strategies to streamline team workflows.

SKILLS

Front-End	React, React Native, TypeScript, JavaScript (ES6+), Angular, Redux, Sass, HTML5/CSS3, Vite, Mantine
Back-End & APIs	Node.js, RESTful APIs, Protobuf, Fetch/Ajax, OAuth2/OIDC, RBAC, JWT
Architecture	Full-Stack Development, Server-Side Config Management, Authentication Architecture, Component Libraries, Design Systems
UI/UX	Wireframing, Mockups, Prototyping, Accessibility (WCAG), Axure RP Pro, Adobe Illustrator

DevOps & Tools	Git, CI/CD Pipelines, Docker, Podman, DataTables, Jira/Confluence/Bitbucket, VS Code, SharePoint
Testing	Manual Testing, Automated Testing, Console Debugging, Regression Analysis
AI Tooling	Claude Code, AI-assisted development workflows

PROJECTS

- **Bridal Shower Jeopardy** — Real-time multiplayer Jeopardy-style game facilitating live buzzer competition across 4 simultaneous mobile clients, built with React/TypeScript, PartyKit (WebSockets), Mantine, and Vite. Demonstrates full-stack architecture with real-time state sync and responsive mobile UI.
- **Goal Tracker (POC) —** React/TypeScript web app leveraging Mantine to track personal goals and progress with a clean, responsive UI. GitHub
- **Personal Site & Portfolio —** React/TypeScript site leveraging Mantine; showcases design and front-end architecture skills. GitHub
- **Mountain Matrix —** Angular/Material app comparing ski resort data; focused on UI design and data visualization. GitHub

AWARDS & RECOGNITION

- AI TechStarter 2025 — Completed 12-week program covering AI/ML with hands-on project-based learning and engagement with industry experts.
- Selected to Attend Grace Hopper Celebration 2024, Philadelphia, PA.
- Northrop Grumman BRAVO! Award — Recognized for exceptional collaboration and extraordinary results in UI development on an accelerated delivery timeline.
- Northrop Grumman BRAVO! Award — Recognized for dedication and major positive program impact delivering a complex logistics feature on an accelerated schedule.

EDUCATION

M.S. in Computer Information Systems (Concentration: Web App Development)
Boston University, 2022

B.B.A. in Operations and Information Management (Minor: Information Technology)
University of Massachusetts Amherst, 2018